

The easiest way to understand **Gaussian Mixture Models (GMM)** is to first compare it with **K-Means**.

Step 1: K-Means vs GMM

Imagine we have customers based on:

Customer Visits/month Spend (\$)

A	2	100
B	3	120
C	4	150
D	10	600
E	11	650
F	12	700

Visually:

Low spenders High spenders

A B C D E F

K-Means

Suppose we ask K-Means for 2 clusters.

It creates a hard boundary:

Cluster 1 = A B C
Cluster 2 = D E F

Every customer belongs **100% to one cluster**.

No uncertainty.

Step 2: What if someone is in between?

Add a new customer:

Customer Visits Spend

G 7 350

Now G sits in the middle.

A B C G D E F

K-Means says

G belongs to Cluster 2

or

G belongs to Cluster 1

Just one answer.

Step 3: How GMM thinks

GMM asks:

"What is the probability that each cluster generated this customer?"

Instead of:

G = Cluster 2

it says:

Cluster Probability

Cluster 1 40%

Cluster 2 60%

So G is:

40% low spender

60% high spender

This is called **soft clustering**.

The Accent Example from the Slide

Suppose someone speaks English.

You hear their accent.

Can you say with certainty:

100% Mumbai

No.

You might think:

70% Mumbai
20% Pune
10% Ahmedabad

That's exactly how GMM works.

Instead of forcing one answer, it assigns probabilities.

Why "Gaussian" ?

A Gaussian is simply a bell curve.

Example:



Suppose:

Cluster 1

Students spend around ₹500/month.

Mean = 500

Bell curve:

300 400 500 600 700
 ^
 most students

Cluster 2

Professionals spend around ₹3000/month.

Mean = 3000

Bell curve:

2000 2500 3000 3500 4000
^
most people

GMM assumes:

Data was generated by multiple bell curves.

One bell curve per cluster.

Why is GMM better than K-Means?

1. Handles overlap

Consider heights.

Women

Average = 160 cm

Men

Average = 175 cm

But some women are 175 cm and some men are 165 cm.

There is overlap.

Women: 150-175

Men: 165-190

A person who is 170 cm tall could belong to either group.

GMM handles this naturally:

170 cm person

60% Women

40% Men

K-Means would force:

Cluster = Men

or

Cluster = Women

2. Can find elliptical clusters

K-Means assumes clusters are roughly circular.

Example:

```
      * * * * *
    * *
  * *
* *
```

A long stretched shape.

K-Means struggles.

GMM can model:

Ellipse

instead of only circles.

That's why the slide says:

Clusters can be stretched ellipses, not just circles.

Real E-Commerce Example

Suppose we cluster customers using:

- Number of visits/month
- Average spend

We may get:

Cluster A

Occasional buyers

Cluster B

Regular buyers

Cluster C

Premium buyers

Customer X:

Visits = 8/month
Spend = ₹2500

GMM might say:

Cluster	Probability
Occasional	10%
Regular	70%
Premium	20%

This is useful because marketing can target him as a likely regular buyer who is moving toward premium status.

One-Line Memory Trick

K-Means

"Which cluster is closest?"

Result:

One cluster only

GMM

"Which bell curve most likely generated this point?"

Result:

Probabilities across clusters

Think of it like this:

- **K-Means = Judge** → one final verdict
- **GMM = Detective** → assigns probabilities to multiple possibilities

That's the core idea behind Gaussian Mixture Models.